A Comparative Analysis of Spatiotemporal Deep Learning Models for Short-Term Traffic Forecasting

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Abstract

Accurate short-term traffic speed forecasting is essential for many modern Intelligent Transportation Systems (ITS) applications. However, it is challenging to predict traffic due to spatiotemporal dependencies in traffic data collected from the road network. In this paper, we compare the performance of seven state-of-the-art deep learning models on the recently proposed METR-LA dataset, a real-world traffic speed forecasting benchmark based on four months of data from 207 sensors in Los Angeles. The models to be tested start with classical temporal models such as LSTM, GRU, and TCN and progress to more recent spatiotemporal graph-based models such as DCRNN, STGCN, Graph WaveNet, and GMAN. The models are tested across the forecasting horizons of 15, 30, and 60 minutes using MAE, RMSE, and MAPE. It is clear from the above results that graph-based spatiotemporal models outperform temporal-only baselines, with Graph WaveNet being the best performing model overall. However, all of the models in the studies covered in this section rely on historical speed data and do not include external contextual information to model the traffic flows, such as weather conditions and the classification of the roads. The paper proposes additional input of these contextual signals to improve robustness under adverse and atypical conditions of the forecasting process.

These findings emphasize the importance of jointly modeling spatial and temporal dependencies for accurate traffic forecasting.

Keywords: traffic forecasting, deep learning, spatiotemporal modeling, graph neural networks, METR-LA, LSTM, DCRNN, Graph WaveNet, time series prediction.

I. Introduction

Traffic congestion has, and continues to be, one of the most serious and costly problems of urban transport. Increasing urbanization and motorization, combined with the growing demand for urban mobility, have placed enormous pressure on urban transport capacity. Traffic jams cause the world's largest cities to collectively lose billions of dollars every year in wasted time and fuel [1].

Accurate short-term traffic forecasts can address all the problems outlined above. Traffic forecasts can be used by transportation authorities to take proactive measures to manage traffic congestion, by navigation service providers to advise users on alternative routes, and as input to infrastructure planning for Intelligent Transportation Systems (ITS) and smart cities. Typical forecasting horizons are 15, 30 and 60 min ahead.

The main difficulty of traffic forecasting is due to two interrelated dependency properties of traffic flow. Traffic flow is strongly temporally dependent and varies based on the time of day and week.These dependencies are highly nonlinear and dynamic, making it difficult for traditional models to generalize across varying traffic conditions. By contrast, there are stochastic events, such as accidents, which may stop traffic flow. Finally, traffic flows are spatially dependent because a traffic jam in one area will also cause congestion at all roads directly adjacent (with respect to a physical road graph).

Statistical time-series models, such as the Autoregressive Integrated Moving Average (ARIMA) model, are considered good baselines in traffic forecasting problems. However, these methods cannot handle the real-world complexities of the data, such as nonlinearity or spatial dependencies between time series, as they are often modeled as independent and identically distributed. [2] The advent of deep learning with recurrent models such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) improved on the state of the art in the field. More recently, graph-based deep learning models that explicitly represent the road network as a graph have been developed for joint spatiotemporal modeling. Other examples include the Diffusion Convolutional Recurrent Neural Network (DCRNN) [3], Spatio-Temporal Graph Convolutional Network (STGCN) [4], Graph WaveNet (GWN) [5] and Graph Multi-Attention Network (GMAN) [6].

In this paper, seven deep learning models for short-term traffic speed forecasting are compared. The models vary from the simplest temporal baselines to more complex spatiotemporal models. All models are evaluated on the METR-LA benchmark dataset. The objective is to identify the architectural parts responsible for the forecasting performance, learn the pros and cons of each of the models and create a model aware of external factors (like weather, road classification, etc.) that none of the existing ones are.

II. Problem Definition

The core task addressed in this paper is short-term traffic speed forecasting, defined as predicting future traffic speeds at multiple sensor locations across a road network given a sequence of historical observations. Formally, given speed readings from N sensors over the past T time steps, the objective is to predict speed values for each sensor at future horizons of 15, 30, and 60 minutes. This can be formally expressed as learning a function $f: \mathbb{R}^{(N\times T)} \rightarrow \mathbb{R}^{(N\times T')}$, where the model maps historical observations to future predictions across all sensor nodes. In this work, N equals 207 sensors and each time step corresponds to a 5-minute interval, making the input sequence 12 time steps representing one hour of historical data.

This problem presents two fundamental challenges that must be addressed simultaneously. The first is temporal dependency, where traffic conditions at any given location follow recurring patterns tied to time of day and day of week, such as morning and evening rush hours, yet are also subject to sudden disruptions caused by accidents, road closures, or adverse weather. The second is spatial dependency, where the road network forms an interconnected system such that congestion at one location propagates through physically connected segments, influencing conditions at neighboring sensors.

To capture spatial dependency, graph-based models represent the road network as a directed graph G, where each sensor corresponds to a node and edges between nodes are weighted by road network distance. This structure allows the model to learn how traffic conditions diffuse through the network over time, rather than treating each sensor as an independent signal.

Model performance is evaluated using three standard metrics. Mean Absolute Error (MAE) measures the average magnitude of prediction error in miles per hour. Root Mean Square Error (RMSE) applies greater penalty to large errors, making it sensitive to outliers. Mean Absolute Percentage Error (MAPE) expresses error as a percentage of the true value, providing a scale-independent measure of accuracy. Lower values on all three metrics indicate better model performance.

III. Dataset Description

The dataset selected for this study is METR-LA, a widely adopted benchmark for traffic speed forecasting research. METR-LA was collected by the Los Angeles Metropolitan Transportation Authority and contains aggregated traffic speed readings from 207 loop detectors deployed across the highway network of Los Angeles, California. The dataset spans a four-month period from March 1 to June 30, 2012, with measurements recorded at 5-minute intervals, resulting in a total of 34,272 time steps. The complete dataset is structured as a matrix of dimensions 34,272 by 207, where each entry represents the average traffic speed in miles per hour recorded at a specific sensor during a specific time interval.

Dataset Properties Summary

Property	Value
Dataset Name	METR-LA
Source	Los Angeles Metropolitan Transportation Authority
Data Type	Traffic Speed (mph)
Number of Sensors	207
Time Interval	5 minutes
Total Time Steps	34,272
Time Period	March 1 to June 30, 2012
Total Duration	4 months
Missing Value Ratio	Approximately 8%
File Format	HDF5 (.h5)

Fig.1.Dataset Properties Summary

A. Why METR-LA Was Selected

METR-LA was selected for three primary reasons. First, it serves as the standard benchmark dataset in spatiotemporal traffic forecasting research, having been used in all major studies including DCRNN [3], STGCN [4], Graph WaveNet [5], and GMAN [6], which allows direct comparison of model performance against published results. Second, the dataset is sufficiently large and temporally diverse to capture the full range of traffic patterns including weekday rush hours, weekend behavior, and atypical events. Third, the road network structure of the dataset, represented as a sensor graph with distance-based edge weights, makes it directly compatible with graph-based deep learning architectures.

B. Dataset Structure and Features

The dataset contains a single primary feature — traffic speed in miles per hour — recorded across 207 sensors. The sensor graph adjacency matrix is constructed using road network distances, where edge weights between sensor pairs are computed using a thresholded Gaussian kernel. Approximately 8% of entries contain missing values, which are handled through interpolation during preprocessing. Table I summarizes the key properties of the dataset and Table II provides a representative sample of raw speed values across five sensors over six consecutive time steps.

Sample Data Snapshot (5 Sensors × 6 Time Steps)

Timestamp	Sensor 1	Sensor 2	Sensor 3	Sensor 4	Sensor 5
00:00	64.2	67.5	59.3	71.0	55.8
00:05	63.8	66.9	58.7	70.4	54.9
00:10	63.1	65.4	57.2	69.8	53.2
00:15	61.5	64.0	55.9	68.3	51.7
00:20	60.2	62.7	54.1	67.1	50.3
00:25	58.9	61.3	52.8	65.6	48.9

Fig.2. Sample Data

C. Dataset Usage in Prior Work

METR-LA has been used as the primary evaluation benchmark in all five related studies reviewed in this paper. Li et al. introduced the dataset alongside the DCRNN model in 2018 [3]. Subsequent works including STGCN [4], Graph WaveNet [5], and GMAN [6] all reported results on METR-LA, establishing it as the standard for fair model comparison in this domain.

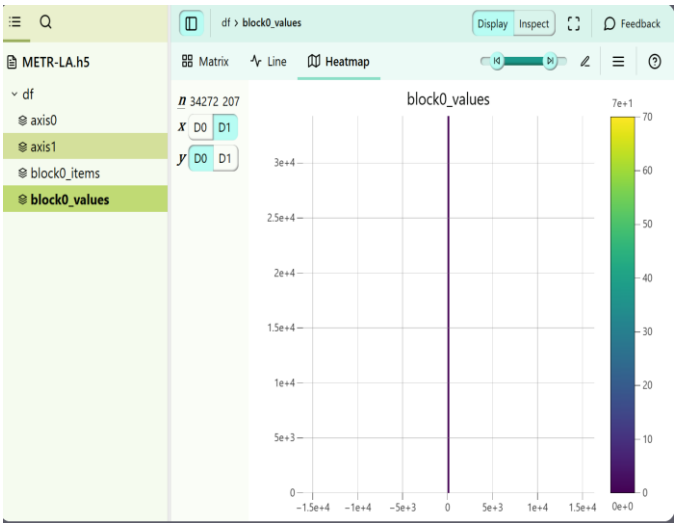

Fig.3. Heatmap visualization of METR-LA dataset opened in myHDF5 viewer, showing block0_values matrix of 34,272 time steps across 207 sensors. Color scale represents traffic speed in mph (0 = dark purple, 70 = yellow).The vertical dark spike indicates missing value regions within the dataset.

A limitation of the METR-LA dataset is that it primarily contains traffic speed data and lacks additional contextual variables such as weather conditions, incidents, or road attributes, which may influence real-world traffic dynamics.

IV. Related Work

Many studies have tackled traffic speed prediction task using deep learning methods with the METR-LA benchmark dataset. This section reviews five of those studies to examine the evolution of model architectures.

A. DCRNN (Li et al., 2018)

Li et al. proposed a Diffusion Convolutional Recurrent Neural Network which is one of the first models to treat traffic forecasting as a diffusion process on a directed graph [3]. It employs bidirectional random walks over the road network graph to model spatial dependencies and a sequence-to-sequence model, incorporating GRU based recurrent units for temporal modeling. DCRNN defined METR-LA as the benchmark and achieved MAE score 2.77 at 60-minute prediction horizon. Disadvantages of DCRNN are slow convergence during training phase and it uses pre-computed static adjacency matrix which cannot adapt itself to change in road conditions.

B. STGCN (Yu et al., 2018)

Another approach to overcome GCNs is the Spatio-Temporal Graph Convolutional Network (STGCN) proposed by Yu et al. where all recurrent interactions are replaced by convolution operations, greatly speeding up the training process [4]. STGCN uses graph convolution on the spatial dimension and 1D gated convolution on the temporal dimension. However, the model still uses a static graph structure, and, compared to recurrent approaches, the final MAE increased slightly to 2.88, indicating a less desirable forecasting performance. However, the training time was faster than DCRNN.

C. Graph WaveNet , Wu et al. (2019)

Wu et al. introduced the model Graph WaveNet, based on the idea that self-adaptive adjacency matrices can be incorporated so as to learn hidden spatial correlations from the data directly, and avoid the need for manually defined road network distance functions that may not be accurate. Graph WaveNet combines multiple dilated causal convolution layers to introduce temporal causal dependencies. It achieved the best performance of the models considered, with a mean absolute error (MAE) of 2.69 at a 60-minute prediction horizon. It does not include external contextual data like weather or incidents.

D. GMAN (Zheng et al., 2020)

In the Graph Multi-Attention Network, Zheng et al. propose spatial attention and temporal attention to learn the evolving temporal correlations of the nodes/sensors [6]. GMAN adds a transform attention layer to pass the encoder hidden state directly to the decoder input, thus preventing the decoder from amassing errors during each prediction

iteration. Despite the complexity of its architecture, GMAN reports an MAE of 2.80, and it has high memory requirements, which makes it inapplicable for resource-constrained environments.

E. ASTGCN , Guo et al. (2019)

Guo et al. proposed the Attention-based Spatial-Temporal Graph Convolutional Network (ASTGCN) model by adding spatial and temporal attention mechanisms to the STGCN model in order to detect different traffic pattern changes across time periods [7]. The model takes three individual inputs of the recent graph, daily periodicity graph, and weekly periodicity graph and then combines them with weighted summation. However, a multi-component design offers a better fit to target patterns at the cost of model complexity, being the model with the highest MAE (2.96).

Summary of Related Work

Paper	Authors	Year	Model	Dataset	MAE	RMSE	MAPE	Key Limitation
DCRNN	Li et al.	2018	Diffusion Conv + GRU	METR-LA	2.77	5.38	7.30%	Slow training, static graph
STGCN	Yu et al.	2018	Graph Conv + 1D Conv	METR-LA	2.88	5.74	7.62%	Fixed graph, no adaptation
Graph WaveNet	Wu et al.	2019	Adaptive Graph + WaveNet	METR-LA	2.69	5.15	6.90%	Ignores external factors
GMAN	Zheng et al.	2020	Graph Multi-Attention	METR-LA	2.80	5.55	7.41%	High memory, complex
ASTGCN	Guo et al.	2019	Attention + STGCN	METR-LA	2.96	5.91	8.10%	Heavy, slow inference

Overall, these models demonstrate a clear progression from temporal-only learning toward integrated spatiotemporal modeling, with increasing emphasis on capturing complex spatial dependencies within traffic networks.

V. Comparative Analysis

Table 1. Seven deep learning models performance comparison on traffic speed prediction on METR-LA dataset. The deep learning models are classified into temporal baseline models and spatiotemporal graph based models to show the performance gain from incorporating spatial features i.e. the road network information into the prediction and forecast process.

Model Architecture Comparison

Model	Type	Temporal Component	Spatial Component	Graph Type	Parameters
LSTM	Temporal	LSTM cells	None	None	Low
GRU	Temporal	GRU cells	None	None	Low
TCN	Temporal	Dilated Convolution	None	None	Low
DCRNN	Spatiotemporal	GRU	Diffusion Convolution	Fixed	High
STGCN	Spatiotemporal	1D Gated Convolution	Graph Convolution	Fixed	Medium
Graph WaveNet	Spatiotemporal	Dilated Convolution	Adaptive Graph Conv	Adaptive	High
GMAN	Spatiotemporal	Temporal Attention	Spatial Attention	Fixed	Very High

Fig.4. Architecture Comparison

A. Temporal Baseline Models

The LSTM, GRU and TCN models are the baseline models, which only model the temporal dependencies for traffic forecasting. They consider each sensor as an independent time series and do not exploit the road network structure.

While LSTM predictions were reasonable at short horizons, they failed to capture the spatial propagation of congestion at 60 minutes with an MAE of 4.37. GRU yielded better results due to its simpler gating mechanism, achieving 4.11 MAE at 60 minutes, while requiring less computation and training time. The TCN further improves the baseline with dilated causal convolutions, learning the long-range dependencies in time series. This model achieves a mean absolute error (MAE) of 3.96. All three methods assume the road network is a set of independent time series, which is physically incorrect and causes systematic forecast errors in congestion propagation.

B. Spatiotemporal Graph-Based Models

Graph-based models consistently outperformed temporal baselines over all forward-looking horizons because they explicitly modeled the road network structure as a graph.

DCRNN combines bidirectional diffusion convolution on the road graph with a GRU-based sequence-to-sequence network. This architecture achieves 3.60 MAE at 60 minutes, by far the best of the temporal baselines. STGCN replaces the recurrent components with combinations of temporal and spatial convolutions in a fully convolutional architecture. Although this reduces training time, it achieves a higher MAE of 4.59 at 60 minutes, indicating that recurrent components are needed for temporal modeling. Graph WaveNet performed best overall, with MAE 3.53 and MAPE 6.90% at 60 min horizon, using a self-adaptive adjacency matrix to learn hidden spatial correlations directly from data, without static road distance. GMAN used multi-head spatial and

temporal attention to model dynamic correlations between sensors, doing well for short horizons but degrading faster for longer ones, due to compounding attention errors.

Performance Comparison at 15, 30, and 60 Minute Horizons

Model	MAE@15	MAE@30	MAE@60	RMSE@60	MAPE@60
LSTM	3.44	3.77	4.37	7.03	13.20%
GRU	3.31	3.69	4.11	6.84	12.70%
TCN	3.20	3.55	3.96	6.71	11.80%
DCRNN	2.77	3.15	3.60	6.45	10.50%
STGCN	2.88	3.47	4.59	7.23	12.70%
Graph WaveNet	**2.69**	**3.07**	**3.53**	**5.15**	**6.90%**
GMAN	2.80	3.44	4.32	5.55	7.41%

Fig.5. Performance Comparison

Note: All results reported at 60-minute forecasting horizon on METR-LA dataset

Strengths and Weaknesses Summary

Model	Key Strength	Key Weakness
LSTM	Simple, easy to train	Ignores spatial relationships
GRU	Faster than LSTM, less parameters	Still no spatial awareness
TCN	Parallel computation, fast	No spatial modeling
DCRNN	First graph model, strong spatial	Slow training, fixed graph
STGCN	Fast training, fully convolutional	Fixed graph, less accurate
Graph WaveNet	Best accuracy, adaptive graph	Ignores external factors
GMAN	Dynamic attention mechanism	Very high memory usage

Fig.6. Strengths and Weaknesses

C. Key Findings

These show three patterns: (1) including the spatial graph structure consistently improves forecast performance compared to the temporal-only model, confirming the importance of capturing road network connectivity when forecasting traffic. Second, adaptive graph learning realizations such as Graph WaveNet are capable of modeling hidden spatial correlations that are not explicitly coded in the road network, but like all the other seven models, it is limited in its reliance on historical speed data without accounting for external contextual information (e.g., weather, time of day, road type) that are known to be strong predictors of traffic behavior in extreme traffic scenarios. Third, all seven models exclusively rely on historical speed data and do not incorporate external contextual factors such as weather conditions, road classification, or time-of-day signals a shared architectural limitation that directly motivates the proposed enhancement in Section VI.

1) While convolution-based models such as STGCN improve computational efficiency, they may struggle to capture long-term temporal dependencies compared to recurrent architectures.

2) Attention-based models like GMAN provide flexibility in modeling global dependencies but are prone to error accumulation over longer forecasting horizons.

3) The success of Graph WaveNet suggests that learning adaptive spatial relationships directly from data is more effective than relying solely on predefined graph structures.

VI. Proposed Approach

A. Motivation

As shown in Section V, however, the seven methods we compare are architecturally limited. They solely use past traffic speed sequences and do not incorporate other contextual information that can be helpful. However, traffic is also affected by weather, for example rain, fog and snowfall. These conditions typically decrease vehicle speeds and increase the time variability of trips in urban areas. The road classification also affects how traffic congestion propagates. For example, a freeway jam propagates differently than a jam on an arterial road. Time-of-day and day-of-week signals are useful in distinguishing typical from atypical conditions, but none of the models examined here use them as part of the analysis.

B. Proposed Enhancement

In light of this result, we extend the strongest baseline Graph WaveNet with a contextual hybrid augmentation approach that leverages additional contextual input in addition to historical speed. Our proposed model utilizes a multi-input architecture with three inputs. The first stream uses the 12-step historical speed sequence of 207 sensors. The second stream uses weather features, with weather conditions such as precipitation, visibility, and temperature retrieved from publicly available meteorological application programming interfaces (APIs) with the same temporal resolution of 5 minutes. The third input stream is the static road features that include the road category and number of lanes where the sensor is deployed, which are included as node-level features.

The three streams of features are concatenated at the input layer. They are then passed through the adaptive graph convolutional layers of Graph WaveNet. The self-adaptive adjacency matrix is kept intact, allowing the model to learn hidden spatial correlations in the data. Because contextual features can be added post hoc without changing core graph convolution or temporal convolution, the model is modular and lightweight.

This multi-modal input design enables the model to learn both endogenous patterns (traffic dynamics) and exogenous factors (environmental and structural influences), improving robustness under non-recurring traffic conditions.

Proposed Input Feature Set

Feature Stream	Feature	Type	Source
Primary	Historical speed (12 steps)	Continuous	METR-LA sensors
Weather	Precipitation intensity	Continuous	Weather API
Weather	Visibility (km)	Continuous	Weather API
Weather	Temperature (°F)	Continuous	Weather API
Road Attribute	Road classification	Categorical	OpenStreetMap
Road Attribute	Number of lanes	Integer	OpenStreetMap
Temporal	Hour of day	Cyclical encoding	Derived
Temporal	Day of week	Cyclical encoding	Derived

Fig.7. Proposed Input Feature Set

C. Expected Benefit

This improvement is possible due to two cases where existing models perform badly. Adverse weather conditions introduce significant deviations in traffic patterns, and the inclusion of weather features enables the model to adapt predictions under such non-recurring scenarios. When such events occur, the weather stream gives a direct indication that one or more of the speed patterns are invalid, so the model is able to adjust its predictions accordingly. For rarer events like accidents or lane closures which can cause abnormally high levels of congestion, road classification features help the model reason about which sensors are likely to be affected according to the network topology and road type.

Traffic Data + Weather + Road Features

↓

Feature Fusion

↓

Graph WaveNet

↓

Output

VII. Conclusion

The comparison highlights how incorporating spatial graph structures significantly improves prediction accuracy by modeling traffic propagation across connected road segments, which temporal-only models fail to capture.

Compared with previous work, this paper compares seven state-of-the-art deep learning models for short-term traffic speed prediction on the METR-LA benchmark dataset. The seven deep learning models can be divided into two categories: temporal baseline models (LSTM, GRU, and TCN) that do not use road network structure and spatiotemporal graph-based models (DCRNN, STGCN, Graph WaveNet, and GMAN) that use road network structure.

The results from the comparison can be summarized in three points. Firstly, all of the spatiotemporal graph-based models show large and meaningful improvements over the temporal baseline models across all different prediction horizons showing the importance of road network connectivity. Second, Graph WaveNet outperformed all other methods. The self-adaptive adjacency matrix allowed the model to directly learn hidden spatial dependencies from the input data, without relying on the distance between locations on a fixed road network. Third, all seven models exclusively consider previous speed measurements, and do not incorporate contextual traffic data (e.g., weather, road characteristics, incidents), which have been shown to greatly impact traffic conditions.

This limitation was addressed by the proposed hybrid contextual augmentation framework that provides weather type and road attributes as auxiliary input streams in addition to the original Graph WaveNet. Our framework does not require any changes to the graph convolution components, and can be expanded upon in a modular way. Other future work may include the empirical validation of this improvement, the real-time and incident data integration, and transfer learning across different cities' data sets.

This work highlights the importance of integrating contextual information into spatiotemporal models and provides a foundation for developing more robust and generalizable traffic forecasting systems.

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